CS143 Spring 2026 – Written Assignment 1

Due Thursday, April 16, 2026 11:59 PM PDT

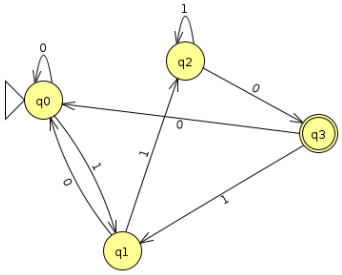
This assignment covers regular languages, finite automata, and lexical analysis. You may discuss this assignment with other students and work on the problems together. However, your write-up should be your own individual work, and you should indicate in your submission who you worked with, if applicable. Assignments can be submitted electronically through Gradescope as a PDF by 11:59 PM PDT on the due date. Please review the course policies for more information: `http://web.stanford.edu/class/cs143/policies/index.html`. A LaTeX template for writing your solutions is available on the course website. To create finite automata diagrams, you can either use the Ti*k*Z package directly by following the examples in the template, or a tool like `https://madebyevan.com/fsm/`.

1. Write regular expressions *and* DFAs that recognize the following languages over the alphabet $\Sigma = \{0, 1\}$. Your DFAs must contain no more than 4 states.

   (a) The set of strings that end with 110.

   Solution:

   Regularni izraz za dati jezik je:
   (0|1)*110

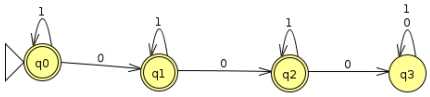
   

   {q0, ..., q3} skup stanja
   q0 - pocetno stanje
   q3 - zavrsno stanje

   (b) The set of strings that contain less than three 0's.

   Solution:

   Regularni izraz za dati jezik je:
   1*0?1*0?1*

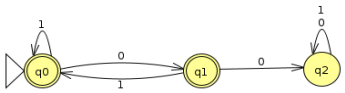
   

   {q0, ..., q3} skup stanja
   q0 - pocetno stanje
   {q0, q1, q2} - zavrsna stanja

(c) The set of strings that do not contain two or more consequent 0's.

Solution:

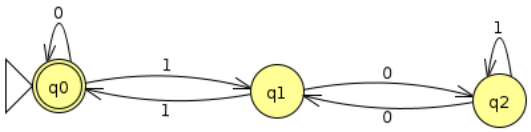
Regularn izraz za dati jezik je:
1*(01+)*0?



{q0, q1, q2} - skup stanja
q0 pocetno stanje
q0, q1 zavrsna stanja

(d) The set of strings that, when interpreted as a binary number, is a multiple of 3 (as an edge case, the empty string shall be interpreted as number 0, which is a multiple of 3). **Hint:** For this subproblem, it might be easier to draw the DFA first, and then construct the regular expression by considering what paths are possible in the DFA to reach the accept state.

Solution:



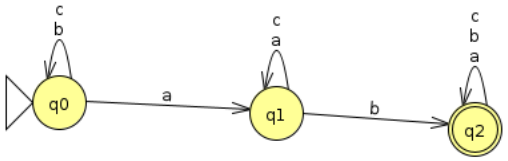
Regulatni izraz koji odgovara
datom dijagramu je:
(0|1(01*0)*1)*

{q0, q1, q2} - skup stanja
q0 - pocetno stanje
q1 - zavrsno stanje
Stanja su redom ostaci pri dijeljenju
datog broja sa 3, za svako stanje se racuna
prethodni_izraz * 2 + trenutni_karakter, shodno
tome reacuna se novi prelaz

2. Write NFAs that recognize the following languages over the alphabet $\Sigma = \{a, b, c\}$.

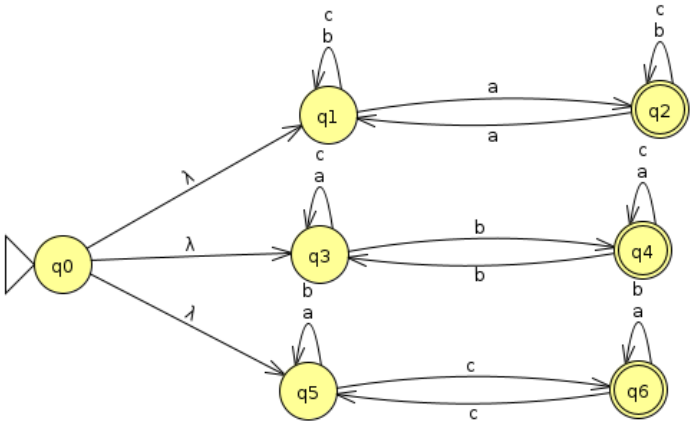
   (a) The language L where a string w is in L iff there exists $i < j$ such that $w[i] =$`'a'` and $w[j] =$`'b'`. For example, `ab`, `bacb` are in L, where `ba`, `ac` are not. Your NFA should have no more than 3 states.

   Solution:



   (b) The language L where a string w is in L iff some character $x \in \{a, b, c\}$ appears an odd number of times in w. For example, `ab` is in L, but `aa` is not. Your NFA should have no more than 8 states.

   Solution:

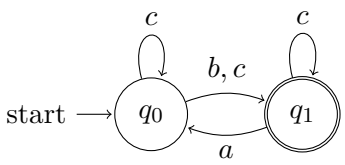


3. Using the techniques covered in class, transform the following NFAs over the alphabet $\{a, b, c\}$ into DFAs. Your DFAs should not have more than 10 states. Note that a DFA must have a transition defined for every state and symbol pair, whereas a NFA need not. You must take this fact into account for your transformations. Hint: Is there a subset of states the NFA transitions to when fed a symbol for which the set of current states has no explicit transition?

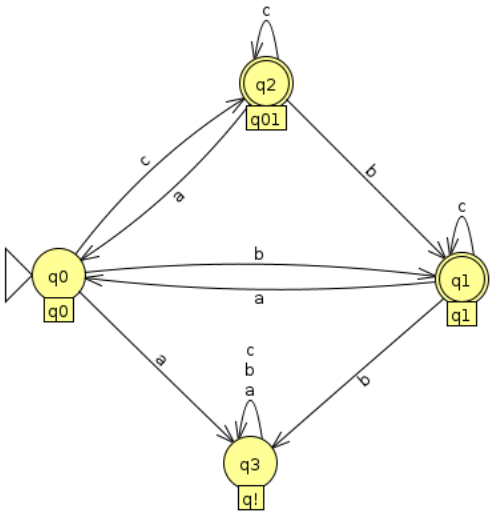
   Also include a mapping from each state of your DFA to the corresponding states of the original NFA. Specifically, a state q of your DFA maps to the set of states Q of the NFA such that an input string stops at q in the DFA if and only if it stops at one of the states in Q in the NFA.

   Tip: for readability, states in the DFA may be labeled according to the set of states they represent in the NFA. For example, state q_{012} in the DFA would correspond to the set of states $\{q_0, q_1, q_2\}$ in the NFA, whereas state q_{13} would correspond to set of states $\{q_1, q_3\}$ in the NFA.

(a)

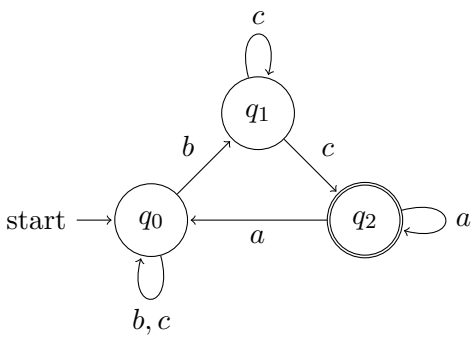


Solution:

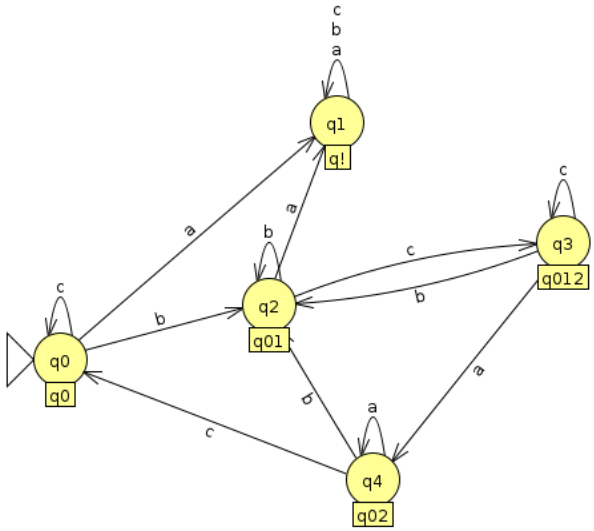


Posmatraju se labele, nisam mogao u jflapu da posedim imena stanja.
Stanje iz DFA koja odgovaraju stanjima u NFA:
q0: {q0}
q1: {q1}
q01: {q0, q1}
q!: {}

(b)

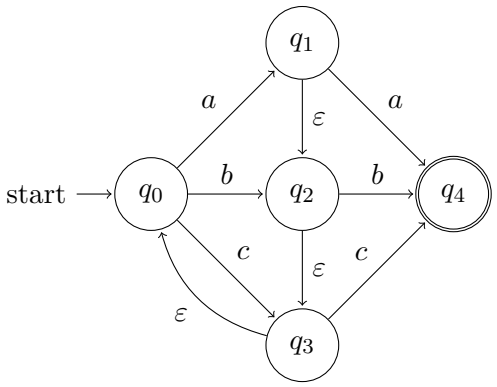


Solution:

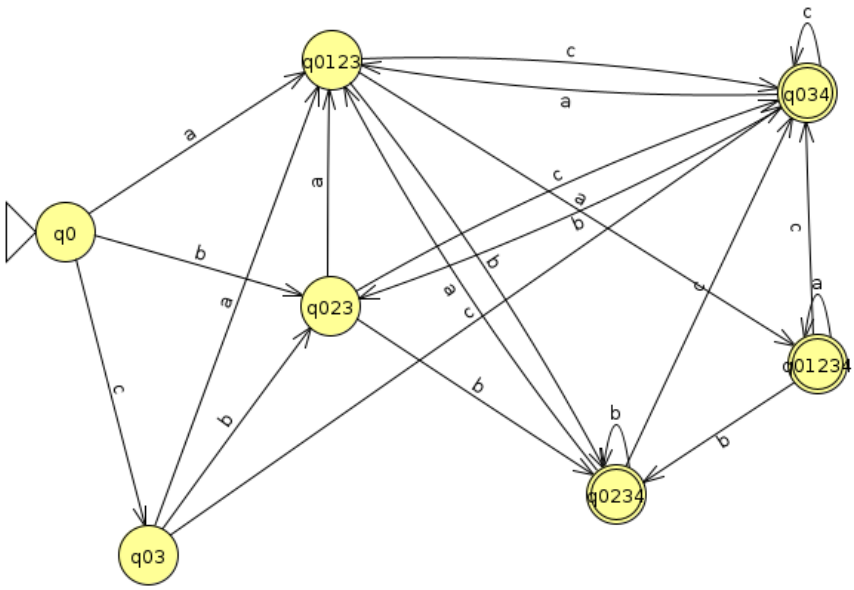


Posmatraju se labele.
Stanje iz DFA koja odgovaraju stanjima u NFA:
q0: {q0}
q01: {q0, q1}
q012: {q0, q1, q2}
q02: {q0, q2}
q!: {}

(c)



Solution:



Stanje iz DFA koja odgovaraju stanjima u NFA:
q0: {q0}
q03: {q0, q3}
q023: {q0, q2, q3}
q0123: {q0, q1, q2, q3}
q01234: {q0, q1, q2, q3, q4}
q0234: {q0, q2, q3, q4}
q034: {q0, q3, q4}

4. Consider the following tokens and their associated regular expressions, given as a **flex** scanner specification:

```
%%
0                       printf("a");
(10|01)                 printf("b");
(101|011+)              printf("c");
```

Give an input to this scanner such that the output lexeme sequence is `aacbbca`.

For ease of grading, please use underlines to show the parts of your input string that correspond to each lexeme, for example, 10 0 0111.

Solution:

Rjesenje je:

0 0 011 01 01 011 0

5. Recall from the lecture that, when using regular expressions to scan an input, we resolve conflicts by taking the largest possible match at any point. That is, if we have the following **flex** scanner specification:

```
%%
do                          { return T_Do; }
[A-Za-z_][A-Za-z0-9_]*      { return T_Identifier; }
```

and we see the input string "`dot`", we will match the second rule and emit T_Identifier for the whole string, not T_Do.

However, it is possible to have a set of regular expressions for which we can tokenize a particular string, but for which taking the largest possible match will fail to break the input into tokens. Give an example of no more than two regular expressions and an input string such that: a) the string can be broken into substrings, where each substring matches one of the regular expressions, b) our usual lexer algorithm, taking the largest match at every step, will fail to break the string in a way in which each piece matches one of the regular expressions. Explain how the string can be tokenized and why taking the largest match won't work in this case.

As a challenge (not necessary for credit), try to find a solution that only uses one regular expression.

Solution:

Uzecu sledeca dva tokena, recimo abcab i abc:

```
%%
abc       { return T_abc ; }
abcab     { return T_abcab ; }
```

Uzecemo i string abcabc, normalna neka tokenizacija bi od ovog stringa uzela dva T_abc tokena, odnosno rastavila bi ovo na abc abc, medjutim, podrazumijevanim ponasanjem u Flexu, preklapanjem najduzeg tokena dobije se abcab kao T_abcab i ostane c koji ne odgovara ni jednom tokenu. Na ovaj nacin se vidim mana ovog greedy pristupa.